PRESENTER GUIDE

How to present this, and what every part of it means

This guide assumes you know nothing about plasma physics. Part 1 explains every idea in the deck from the beginning. Part 2 gives you the words to say, slide by slide. Part 3 is the questions the judges are likely to ask, with honest answers. Part 4 is a short list of words that must not be got wrong.

Read Parts 1 and 2 twice. Once you understand it, the script in Part 2 will sound like your own thinking rather than something memorised.

The one sentence version

We worked out how to measure something that cannot be measured, by watching the light a plasma gives off and using physics to work backwards. We tested it honestly, it failed the first time, we found out why, and we fixed it.

PART 1

The background, explained from zero

1. What is a plasma?

You know solids, liquids and gases. A plasma is the fourth one. If you keep adding energy to a gas, the atoms start to break apart. Each atom is made of a heavy positive centre with light negative electrons around it. Add enough energy and some of those electrons get knocked loose.

What you are left with is a soup of two things mixed together: free electrons, which are negative and very light, and ions, which are the leftover atoms, positive and heavy. That soup is a plasma.

You have seen plasma. Lightning is plasma. The glow inside a neon sign or a fluorescent tube is plasma. The sun is plasma. Most of the visible universe is plasma.

Overall a plasma is electrically neutral, because there are as many negative charges as positive ones. That matters for the next section.

Why industry cares about plasma

Chip factories. Plasma is used to cut microscopic patterns into silicon wafers. Almost every chip in every phone was made using plasma.

Satellites. Electric thrusters push a satellite along by throwing plasma out of the back at very high speed.

Fusion reactors. Fusion works by holding an extremely hot plasma and getting the atoms in it to join together and release energy.

2. What is a sheath?

This is the central idea in our work, so take it slowly.

Put a plasma next to a solid wall. Something happens automatically, without anyone doing anything. The electrons are far lighter than the ions, so they move much faster. They reach the wall first and stick to it. The wall builds up negative charge.

A negatively charged wall pushes other electrons away and pulls positive ions towards it. So very close to the wall you end up with a thin region where the charges are no longer balanced. There are more positive ions than electrons there.

That thin unbalanced layer is called the sheath. In our test setup it is about 0.89 millimetres thick, which is about the thickness of a fingernail.

Because the charges are unbalanced, there is a strong electric force across the sheath. Ions crossing it get accelerated, like a ball rolling down a steep hill. They arrive at the wall moving much faster than they were in the main body of the plasma.

The important consequence: how hard the ions hit the wall is decided by the sheath, not by how hot the plasma is. So if you want to know what the wall is suffering, the sheath is what you have to understand.

3. What is ion energy flux?

This is the number our whole project is about. It has a symbol, which you will see on slide 3: the Greek letter gamma with a small E, written as Γ_E.

In plain words: **ion energy flux is how much energy the ions dump onto each square metre of the wall, every second.**

It is measured in kilowatts per square metre, written kW/m². That is the same kind of unit as a household heater. At our test point the true value is 9.33 kW/m². For a feel: a small room heater is about 1 to 2 kilowatts, so this is roughly like pressing five or six heaters against every square metre of the wall.

Heat	The wall gets hot. That decides how much cooling the machine needs, and so how much power it can be run at.
Wear	The arriving ions knock atoms out of the wall, one at a time. Over months and years this eats the wall away. It is the main reason plasma equipment eventually has to be replaced.
Waste	Energy that ends up heating the wall is energy that did not do the useful job, whether that job was pushing a satellite or etching a wafer.

4. Why can nobody measure it?

There are two ways you could try, and both fail.

Way one: put a probe in

You can push a small metal probe into the plasma and measure the electrical current flowing into it. The problem is that the probe is itself a solid object. As soon as it goes in, the plasma forms a sheath around the probe too. So the thing you are trying to measure is changed by the act of measuring it.

There is a second problem. Near a wall the conditions are harsh, and probes do not survive well there.

Way two: look at the light

A plasma glows. You can point instruments at it from outside, through a window, and learn things without touching it at all. This does not disturb anything. The problem is that each instrument only tells you one piece of the story, and none of them tells you the energy flux.

The four instruments we use, and what each one actually tells you

Emission spectroscopy. The plasma glows. Split that glow into its colours, like a rainbow through a prism. Which colours are bright tells you about the electrons.

Laser-induced fluorescence. Shine a laser in. Some ions absorb the light and re-emit it. Because moving things shift the colour slightly, the shift tells you how fast the ions are travelling.

Thomson scattering. Fire a laser in and look at the light that bounces off the electrons. How much the returned colours are spread out tells you how hot the electrons are.

Interferometry. Send a light beam through the plasma and compare it with a beam that went around. The difference tells you how dense the plasma is.

None of those four is the ion energy flux. And you cannot simply add them together to get it either. That is the gap our work fills.

5. What is an inverse problem?

A **forward** problem is: I know the cause, so what is the effect? If I know exactly what the plasma is doing, I can calculate what each of the four instruments would read.

An **inverse** problem is the other direction: I can see the effect, so what was the cause? I have four instrument readings. What plasma would have produced them?

A everyday comparison. If someone hits a drum in the next room, you hear a sound. Working out what sound a given drum makes is the forward problem. Hearing the sound and working out the size and shape of the drum is the inverse problem. The second one is much harder, because several different drums might sound similar.

That is exactly our situation, and it is why the answer cannot just be read off. We have to search for the plasma state that best explains all four readings at once.

6. What is a forward model, and what do we actually do?

A forward model is a piece of software that answers the forward question. Give it a guess about the plasma, and it predicts what all four instruments would read.

Once you have that, the method is simple to describe:

How the method works, in four steps

1. Guess a plasma state.
2. Use the forward model to predict what the four instruments would read.
3. Compare those predictions with what the instruments actually read.
4. Adjust the guess and repeat, until prediction and reality agree.

When they agree, the guess you are holding is your answer, and the ion energy flux can be calculated from it.

7. Why do we give a range instead of a single number?

A single number with no error bar is not very useful, and it can be dishonest. Our method returns a range together with how confident it is. For example: the flux is about 9.3 kW/m², and we are 90 per cent sure the true value lies between 8.9 and 9.7.

That range is called a **credible interval**. Ninety per cent is the confidence level we chose to report.

And what is coverage?

Coverage is the test of whether that confidence is honest. It is the most important idea in our results, so make sure it is clear in your head.

If you claim ninety per cent confidence, and you run the test ten times, then the true answer should fall inside your stated range about nine times out of ten. If it does, your confidence was honest. If it only falls inside three times, you were overconfident and your error bars were lying.

Our first attempt scored zero out of ten. The true answer fell outside our stated range every single time. That is the worst possible result, because it means we were not just wrong, we were wrong while sounding certain.

8. What is an inverse crime? (this is why we have three tiers)

Imagine you write your own exam paper, write your own answer key, and then mark your own exam using that key. You will score full marks, and it will prove nothing.

Software can make exactly that mistake. If you use one program to create fake test data, and then use the same program to solve it, you get an almost perfect answer that means nothing at all. In this field that mistake has a name: an **inverse crime**. It is the most common way work like ours gets thrown out.

We guard against it by grading ourselves in three tiers. Two things need explaining first: the levels, and then the tiers.

The three levels of physics (L0, L1, L2)

We can calculate the sheath physics in three different ways, from rough and fast to detailed and slow.

L0, analytic	Pen-and-paper formulas. Fast, and approximate.
L1, fluid	Treats the plasma like a flowing liquid and solves it numerically on a computer. Slower, and more realistic.
L2, kinetic	Tracks individual particles one by one. Slowest, and most realistic.

The three tiers of test (T0, T1, T2)

T0, easy	Same level of physics used to make the test data and to solve it, and no noise added. This only checks that the code is not broken. We scored 0.012 per cent error.
T1, medium	Same level on both sides, but with realistic measurement noise added. An optimistic best case. We scored 0.050 per cent.
T2, hard	A **different** level on each side. The test data is made using L1, the fluid calculation, and the solver is only allowed to use L0, the simple formulas. The solver is working with a deliberately imperfect picture of the physics, which is what real life is like. We scored 3.99 per cent, and this is the only number we quote.

Two rules we built into the code so we could not cheat by accident

The camera never sees the plasma. Our model of the detector is only allowed to receive light. It is not allowed to look up the plasma density, because a real camera could not do that either.

One piece of code, not two. The part that makes the test data and the part that solves it share the same physics code. If they were written separately they would slowly drift apart, and results would become meaningless without anyone noticing.

9. What actually happened in our results

This is the story the deck tells. Learn it as a story, not as numbers.

First attempt	We combined all four instruments in the obvious way, treating each as equally trustworthy. The answer was 25.7 per cent away from the truth, and the error bar contained the truth zero times out of ten.
The diagnosis	We tested each instrument on its own, deliberately without looking at the combined result, so we could not fool ourselves. We had written down beforehand that we expected all four to be a little overconfident. We were wrong. Only one was.
The culprit	Emission spectroscopy. On density it claimed to be accurate to about 1 per cent but actually scattered by 30 per cent, so it was about 29 times too sure of itself. On temperature it claimed about 1.5 per cent and scattered by 210 per cent, about 138 times too sure.
The fix	We told the method to trust that one instrument less. Error went from 25.7 per cent to 3.99 per cent, and coverage went from zero out of ten to eight out of ten. All four instruments are still used.
The check	We varied how much we turned that instrument down, across the whole range our two independent estimates disagreed over, a factor of 23. The conclusion did not change. So this is a real diagnosis, not a number we tuned until it looked good.

10. What the numbers on the title slide mean

Accuracy, 96.0%	Our answer is on average 3.99 per cent away from the true value. So we get it right to within about 96 per cent. In physical terms: the true flux is 9.33 kW/m² and we land within about 0.37 kW/m² of it.
Tolerance, ± 2.76%	We ran the whole test ten times with different random noise. The results varied by about 2.76 per cent around that average. It is a measure of how repeatable we are.
Confidence, 90%	The confidence level of the range we quote. We are saying we are 90 per cent sure the truth is inside our stated range.
Coverage, 8 of 10	Out of ten runs, the truth actually fell inside our stated range eight times. That is close to the nine we would want, which means our error bars are roughly honest.

11. What is the readiness level we mention at the end?

NASA has a scale from 1 to 9 for how ready a technology is. Level 1 is a basic idea on paper. Level 9 is something already working in space. It is used worldwide so that funders can compare very different projects fairly.

For software, level 2 has two requirements that count as evidence: the code has to actually exist, and it has to have been tested on simulated data. **We have now met both.**

Level 3 requires testing against real laboratory data. We have not done that, and we say so. That is precisely what we are asking for.

PART 2

The business side, explained from zero

Judges will ask who would pay for this. This part explains every number on the market slide, where it came from, and what we deliberately refuse to claim.

12. What are we actually selling?

Software. Specifically, the program that takes readings from optical instruments and works out the ion energy flux, with an honest error bar attached.

In practice we would not sell it alone. We would sell it together with the optical hardware it needs, because that is what buyers in this field are used to purchasing, and because software on its own has a weaker record here.

Three ways this could earn money

Run the measurement as a service. A thruster company is qualifying a new engine. We come in, take the measurement, hand them the numbers. They pay per job.

Licence the software. A laboratory or factory buys the right to run it themselves, usually priced per user per year.

Research licences. Cheaper versions for universities and national laboratories.

13. Why is there no market report for this?

Normally a founder quotes an analyst report: this market is worth so many billions and growing at so many per cent. We cannot do that, and it is worth understanding why.

Eight of the large market-research firms were checked by name: Grand View, MarketsandMarkets, Precedence, Global Market Insights, Mordor, Fortune Business Insights, Allied and IMARC. **None of them publishes a report on plasma diagnostics.** The category does not exist as far as the analyst industry is concerned.

That cuts two ways, and you should say both.

The bad side	There is no analyst report to point at, and no budget line that a customer already has labelled for this. We would be creating the category, which is harder and slower.
The good side	No competitor has been sized, funded and aimed at this category either. Nobody has taken the position.

There was a temptation here we deliberately avoided. It would have been easy to quote the semiconductor equipment market, which is worth over a hundred billion dollars, and imply that is our market. That is called adjacent-market substitution and it is the single most common way pitches like ours get taken apart. We do not do it, and saying so out loud is worth more than the bigger number would have been.

14. So how did we get to 84 customers?

We counted them, one at a time, by name.

46 private fusion companies	Companies actually building fusion machines. Taken from the Fusion Industry Association membership list.

21 research partners	Universities and national laboratories on the same list.
10 plasma etch tool makers	The companies that build the plasma machines used in chip factories. This is a small and well-known list: Lam Research, Applied Materials, Tokyo Electron, Hitachi High-Tech, SPTS/KLA, Oxford Instruments, Plasma-Therm, Samco, AMEC and NAURA.
7 propulsion test facilities	Places that test electric thrusters in vacuum chambers, including NASA Glenn, NASA JPL, Georgia Tech, the University of Michigan and QinetiQ.

46 plus 21 plus 10 plus 7 is 84.

Why the counting method matters

The fusion membership list is a live web page, and live pages change. Counting it on different days gives slightly different answers, which would make our number impossible to check.

So we do not count the live page. We count a **dated archived copy from 16 May 2026**, stored in the Internet Archive. Anyone in the room can open that exact snapshot and count for themselves and get the same answer. That is the difference between a number you assert and a number you can defend.

Be ready for this. Two honest weaknesses are written into our own report. First, we counted members by their logos on the page, so any member listed without a logo would be missed. Second, of the seven propulsion facilities, only one was confirmed by opening that facility's own web page; the other six came from secondary sources. If asked, say both plainly. They are in our report because we put them there.

15. What do these customers already spend?

A market only exists if money is already moving. It is.

We looked up actual United States government purchase records for one company that sells the physical probes our software would replace. **Seven awarded contracts, ranging from 16,240 dollars to 86,080 dollars per instrument.** These are real published records, not estimates.

That tells us two useful things. The budget line already exists, so nobody has to invent one. And it tells us roughly what an instrument in this field sells for.

16. The four figures along the bottom of the market slide

These size the industries our customers sit in. **They are not our market**, and you must say so, because a judge who thinks you are claiming them will stop believing you.

$4.48B	Raised by private fusion companies in the twelve months to July 2026. On top of 13.26 billion dollars of private money already invested since 2021. Source: the Fusion Industry Association's own annual report.
$133B to $156B	Total semiconductor equipment sales, 2025 rising to a 2027 forecast. Source: SEMI, which is the industry's own trade body and surveys its members directly, so it is better evidence than a market-research firm.
$415M	Authorised through 2027 for the United States fusion milestone programme. Government money committed to this area.

60 instruments	The number of diagnostic instruments planned for ITER, the international fusion reactor being built in France, measuring 101 separate quantities. This is the one to point at if someone asks how much demand there is for diagnostics: a single facility fields sixty of them.

The line to say is: those are the industries our customers sit in, not our own market. What we sell into is the 84.

17. Would licensing alone build a company?

No, and our own report says so. This is worth knowing because it shows you have actually done the arithmetic rather than assumed.

The only published price we could find for comparable scientific software is the University of Cambridge's rate for a package called COMSOL: 3,025 pounds per user per year. If all 84 organisations bought one licence at that rate, that is about 254,000 pounds a year. Three licences each gets to about 762,000 pounds.

That is roughly the size of a single research grant. It is real money, but it is not a business on its own. That is why the plan pairs the software with hardware and with paid measurement work, rather than relying on licences.

18. Has anyone proved there is money in this?

Yes, and it is the strongest commercial fact we have. **Applied Materials, one of the largest semiconductor equipment companies in the world, bought a company called Impedans**, which makes the physical probe our software is designed to replace. The price was not disclosed.

Someone with far better information than us decided this measurement was worth owning. We did not have to guess at that.

Two other sales in the same area, for scale: Oxford Instruments bought Andor for 158.1 million pounds, and Nova bought ancosys for about 90 million dollars. Most sales in this field have been an instrument with software attached, rather than software alone, which is why we plan to sell the pair.

19. What is the readiness level, and why does it matter?

NASA has a scale from 1 to 9 describing how ready a technology is. It is used around the world so that funders can compare very different projects fairly.

Level 1	A basic idea, written down.
Level 2	For software: the code exists, and it has been tested on simulated data. **We have now met both of these.**
Level 3	Tested against real laboratory data. We have not done this, and we say so.
Levels 4 to 9	Increasingly realistic environments, ending with something already working in space.

Being honest about sitting at level 2 is deliberate. Our own commercialization assessment originally rated the project lower than the team's own estimate, and we kept the lower rating. Funders see inflated readiness claims constantly, and a team that rates itself accurately is unusual.

20. What are we asking for, and why those three things?

A plasma source with optical windows	We need a real plasma to point instruments at, and it needs windows so we can look in without touching it.
A reference probe	To check our answer. We need something that measures the same region a different way, so we can compare.
Incubation and mentorship	Guidance, and the institutional base to do this properly. That is what the π-Hub exists for.

Note what we are not asking for: a large amount of money. The next step is access, not cash. For scale, comparable publicly funded projects doing this kind of work receive between 300,000 and 400,000 dollars.

One thing to know if it comes up: the American SBIR grants often used as a benchmark for early-stage work like this are **not available to us**, because they require the work to be performed in the United States. The realistic routes are Indian schemes such as the Startup India Seed Fund and NIDHI-PRAYAS.

21. The two documents behind this

The technical paper	34 pages. Describes the framework, the tests, the results and the three defects the system found in itself. This is where every technical number in the deck comes from.
The commercialization assessment	A separate report on whether this could become a business. It was reviewed five times by independent reviewers who were told to attack it. Every market number in the deck comes from it.

If a judge wants more detail than the deck gives, offer the relevant document. Do not try to recall a number you are unsure of.

PART 3

The script, slide by slide

Times are a guide. The whole thing should run about eight minutes, which leaves room for the introduction and for going slightly slower than planned. Do not rush.

Text in the larger serif type is what you say. The notes around it are for you.

SLIDE 1 · about 15 seconds

Title

Do not read the four boxes out. They are there for the judges to see. You will explain them properly on slide 5.

Good morning. We are Team Astrophage. Our project is about measuring how much energy is delivered to a surface inside a plasma machine, at the exact place where the plasma touches the wall. It is a number that matters a great deal, and at the moment nobody can measure it. We built software that works it out instead, and today we will show you how we tested it, including the part where it failed.

SLIDE 2 · about 45 seconds

The problem

A plasma is a gas with so much energy that the atoms have broken apart into loose electrons and heavy ions. Wherever a plasma touches a solid surface, the ions are pulled into that surface and hit it.

How hard they hit decides three things. The wall gets hot, which limits how much power the machine can run at. The wall slowly wears away, which is the main reason this equipment eventually fails. And any energy that ends up in the wall is energy that did not do the useful work.

Three industries live with this. Satellite thrusters, where the engine slowly eats its own walls. Chip factories, where the energy of the ions decides how cleanly a circuit pattern is cut into silicon. And fusion reactors, where the inner wall has to absorb everything the plasma throws at it.

All three need this number. None of them can measure it.

Pause after the last line. It is the setup for the next slide.

SLIDE 3 · about 55 seconds

The gap

Do not read the formula out loud. Point at it and say what it means.

This is the quantity. It is the energy the ions deliver to each square metre of wall every second. The reason it is hard is in the definition itself: it is not a property you can point an instrument at. It has to be worked out.

There are two ways you might try. You can push a probe in, but the probe is a solid object, so the plasma immediately forms the same kind of layer around the probe. You change the thing you are

trying to measure.

Or you can look at the light, from outside, through a window, touching nothing. That is what these four instruments do. One looks at the colours the plasma gives off. One uses a laser to see how fast the ions are moving. One uses a laser to see how hot the electrons are. One measures how dense the plasma is.

None of them gives us the energy flux, and no combination of them gives it either. So we work backwards. We measure what light can tell us, and use the physics to reconstruct the rest.

SLIDE 4 · about 80 seconds

How we tested it

This is the slide that decides whether a researcher believes you. Slow down.

The obvious danger with work like this is testing yourself against yourself. If you write one program to make the test data, and use the same program to solve it, you get a perfect score that proves nothing. In our field that mistake has a name, and it is the usual reason work like this gets thrown out.

So we did it differently. We create a plasma in software, we know exactly what it is, and then we lock that answer away. We push it through the four stages on the left: the sheath physics, the light it produces, the optics, and the camera. What comes out is a set of measurements, exactly what a real instrument would have seen.

Our solver is then handed only those measurements. It has no way of reaching the answer. That is enforced in the code, not by us being careful. Only at the very end do we unlock the answer and compare.

We can calculate the physics three ways: simple formulas, a fluid calculation, and a full particle simulation. And we grade ourselves in three tiers. The easy tier uses the same physics on both sides with no noise, and just checks the code is not broken. The medium tier adds noise. The hard tier uses different physics on each side: the test data comes from the fluid calculation, and the solver is only allowed the simple formulas.

The hard tier is the only number we quote anywhere, and it is 3.99 per cent.

SLIDE 5 · about 90 seconds

The result

The most important slide. Let the pictures do the work, and pause between the two halves.

The first time we combined all four instruments, we treated them as equally trustworthy. That is the top row. The answer came out 25.7 per cent away from the truth.

But look at the right hand side. We had also stated how confident we were. Out of ten runs, the true answer fell inside our stated range zero times. Not once.

That is worse than simply being inaccurate. It means we were wrong and sounded certain about it, and that is exactly the failure this system exists to catch.

So we checked each instrument on its own, deliberately without looking at the combined result. We had written down in advance that we expected all four to be slightly overconfident. We were wrong. Only one was. Emission spectroscopy was claiming to be accurate to about one per cent when it was really scattering by thirty.

We told the method to trust that one instrument less. The error dropped to 3.99 per cent and the coverage went from zero out of ten to eight out of ten, with all four instruments still in use.

Say this next line clearly. Do not skip it, and do not apologise while saying it. It is the line that earns trust with a researcher.

One thing I want to be clear about. Every measurement behind these numbers is simulated. This shows the method is sound. It does not yet show that it works on real hardware, and that is the next step.

SLIDE 6 · about 60 seconds

The market

There is no analyst report for a plasma diagnostics market. We checked eight research houses and none of them publishes one. So instead of quoting a market size we do not believe, we counted the buyers by hand.

Forty-six private fusion companies. Twenty-one fusion research partners. Ten companies that make plasma etching tools. Seven electric propulsion test facilities. Eighty-four organisations, and we can name every one of them. The count comes from a dated archived snapshot, so anyone here can re-open it and check.

That is a short list, and we think that is a strength. There is nothing to advertise and no leads to generate. Two people could contact this entire market in a month. And they are already buying: seven United States government contracts for a single plasma diagnostic vendor ran from sixteen thousand to eighty-six thousand dollars per instrument, so the budget line already exists.

The figures along the bottom are the industries our customers sit in. Fusion companies raised four point four eight billion dollars in the year to July. Semiconductor equipment sales are going from a hundred and thirty-three billion to a forecast hundred and fifty-six billion. Those are not our market. They are the weather around it. What we sell into is the eighty-four.

SLIDE 7 · about 65 seconds

Where we stand

Five phases. The first is building the framework and testing it against answers we already know. That is done, and it is what we have shown you today. The second is putting real optics on a real plasma source, and that is where we are now. Then first measurements on a real plasma, then checking our answer against a physical probe, then scaling up.

On NASA's readiness scale for software, we have now met both evidence requirements at level two. The code exists, and it has been tested on simulated data. Level three needs real laboratory data, and we have not attempted it yet.

So what we are asking for is specific. Access to a plasma source with optical windows. A reference probe so we can check our answer against a physical measurement. And incubation and mentorship here at the π-Hub.

Thank you. We are happy to take questions.

PART 4

Questions they will ask

Answer briefly and stop. If you do not know, say so. Guessing in front of a researcher is far worse than saying you will follow up.

Is this just machine learning with extra steps?

No. There is no training data and no learned model. We use the actual physics of the sheath as the forward model, and we search for the plasma state that best explains the measurements. That is why it still works in situations we have never seen.

You said it is all simulated. So why should we believe it works?

You should not believe it works on hardware yet, and we do not claim that. What we have shown is that the method is mathematically sound and numerically accurate under stated assumptions, and that it degrades in a measurable way as those assumptions are broken. Real hardware is the next phase and it is what we are asking for.

Three point nine nine per cent does not sound very accurate.

It is an average error of about four per cent on a quantity nobody can measure at all today. And the number we care about more is the coverage: when we say we are ninety per cent confident, the truth really does land inside our range eight times out of ten. A wide honest error bar is more useful than a narrow one that lies.

Why did you show us the failure? Most teams would have hidden it.

Because finding it is the whole point of the system. A method that is wrong is a nuisance. A method that is wrong and confident is dangerous, because you would act on it. Our first result was the second kind, and the machinery caught it.

How do we know you did not just tune the weighting until the number looked good?

Two reasons. The per-channel audit was run separately and deliberately without looking at the combined result. And we varied the weighting across the whole range our two independent estimates disagreed over, a factor of twenty-three, and the conclusion did not change.

What happens if the physics in your solver is wrong?

That is exactly what the hard tier tests. The test data is generated with one level of physics and the solver is only allowed a simpler one, so it is always working with an imperfect picture. The 3.99 per cent already includes that penalty.

Who is your first customer?

The most likely first customer is an electric propulsion group that needs to qualify a thruster, because they currently answer this question by running the thruster for thousands of hours and looking at the wear afterwards.

How much money do you need?

For the next phase we are asking for access rather than cash: a plasma source with optical windows, and a reference probe. Comparable publicly funded projects on this kind of work run in the region of three hundred thousand to four hundred thousand dollars.

What stops a large company doing this themselves?

Nothing stops them, and we should be honest that a well-resourced laboratory could assemble something similar. What we have is a working, tested framework now, and a set of design decisions that took a lot of care to get right.

Where does the number eighty-four come from?

We counted them by name. Forty-six fusion companies and twenty-one research partners from an archived copy of the Fusion Industry Association membership list dated the sixteenth of May, ten plasma etch tool manufacturers, and seven propulsion test facilities. We use the archived copy rather than the live page so that the count can be reproduced by anyone.

Is eighty-four customers not far too small a market?

It is small, and it is real. Every one of them is a facility spending millions on equipment, and they are already buying instruments in this area at between sixteen and eighty-six thousand dollars each. A short named list also means almost no cost to reach them. We would rather show you a number you can check than a market size we cannot defend.

Why not just tell us the semiconductor market size?

Because we would not be entitled to it. Quoting a hundred-billion-dollar equipment market and implying it is our addressable market is the most common mistake in pitches like this one. Those figures are the industries our customers sit in, not our market.

Could you not just licence the software and be done?

We did the arithmetic and it does not work on its own. At the only published comparable price we could find, licences across all eighty-four organisations come to roughly the size of a single research grant. That is why we pair the software with hardware and with paid measurement work.

Has anyone shown there is real commercial value here?

Applied Materials bought Impedans, the company that makes the physical probe our software replaces. The price was not disclosed, but someone with much better market information than us decided this measurement was worth owning.

How solid is your customer count, really?

Two weaknesses are written into our own report. We counted members by their logos, so a member listed without one would be missed. And of the seven propulsion facilities, only one was confirmed by opening that facility's own page. We would rather tell you that than have you find it.

Can we see the code?

The licensing decision is still open, so it is not public yet. We can walk you through the architecture and the test results in detail.

PART 5

Words that must not be got wrong

These are small distinctions, and a researcher in the room will notice every one of them. Learn the right-hand column.

Accuracy, not error	The error is 3.99 per cent, so the accuracy is 96 per cent. Do not say the accuracy is 3.99 per cent. It sounds like the method barely works.
Verified, not validated	We have verified the method against simulated data. Validated means tested against real hardware, and we have not done that. Using the wrong word here will be picked up.
Simulated, not real	Say the word simulated at least once, clearly, on slide 5. Volunteering it is a strength. Being caught not saying it is fatal.
Coverage, not confidence	Confidence is what we claim, ninety per cent. Coverage is what actually happened, eight out of ten. They are different things and the second one is the real test.
Sheath, not surface	The sheath is the thin charged layer next to the wall. It is not the wall itself.
Ions, not atoms	In a plasma the relevant particles hitting the wall are ions, which are atoms that have lost an electron.

If you are asked something you do not know

Say this: *“I do not have that number in front of me, but it is in our paper and I will follow up.”* Then stop talking. That answer costs you nothing. A wrong number given confidently costs you the whole presentation.

The last thing to remember

The strongest thing about this project is not the 3.99 per cent. It is that you tested yourself honestly, you failed, you found out exactly why, and you fixed it, and then you told people about it. Most teams in the room will present only the parts that worked. Say the failure out loud, calmly, as a result rather than as an apology. That is what will be remembered.